

Lab Report: Word Frequency Analyzer Using C++ STL

Title

Word Frequency Analyzer Using C++ STL

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Course: C++ programming

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Objective

- Read text from an input file.
- Remove punctuation and convert words to lowercase.
- Ignore common stop words and short words (length ≤ 3).
- Count the frequency of each word.
- Display and save words that appear more than two times.

Introduction

This program performs basic text analysis using C++ Standard Template Library (STL). It reads a text file, cleans each word, removes common stop words, counts word frequencies using a map, and writes the results to an output file.

Libraries Used

- `<iostream>` – Input and output

- <fstream> – File handling
- <map> – Store word frequencies
- <set> – Store stop words
- <string> – String operations
- <cctype> – Character functions

Algorithm

1. Read the input and output file names.
2. Open the input file.
3. Read each word from the file.
4. Remove punctuation and convert to lowercase.
5. Ignore stop words and words with length ≤ 3 .
6. Count the frequency of each remaining word using a map.
7. Display and save words with frequency greater than 2.
8. Close the files.

Sample Input

Data Science is interesting.
Data helps in Machine Learning.
Machine Learning is useful.
Data Machine Machine.

Sample Output

Words with frequency greater than 2:

data : 3

machine : 3

Features

- Reads text from a file.
- Removes punctuation.
- Converts words to lowercase.
- Ignores stop words.
- Counts word frequencies.
- Saves output to a file.
- Handles file opening errors.

Advantages

- Easy to understand and implement.
- Efficient use of STL containers (map and set).
- Useful for basic text analysis and preprocessing.

Limitations

- Only alphabetic characters are considered.
- Uses a limited list of stop words.
- Ignores numbers and special characters.

Conclusion

This program successfully analyzes a text file by counting the frequency of meaningful words. It demonstrates the use of C++ STL containers, file handling, and string processing, making it a simple and effective example of text analysis.

You, 2 days ago | 1 author (You)

```
#include <iostream>
#include <fstream>
#include <map>
#include <set>
#include <string>
#include <cctype>
```

```
using namespace std;
```

```
// Convert word to lowercase and remove punctuation
```

```
string cleanWord(string word) {
    string cleaned = "";

    for (char ch : word) {
        if (isalpha(ch)) {
            cleaned += tolower(ch);
        }
    }

    return cleaned;
}
```

You, last week • Words Count Program

```
int main() {
    string inputFileName, outputFileName;
    ifstream inputFile;
    ofstream outputFile;

    map<string, int> wordFrequency;

    set<string> stopWords = {
```

```
int main() {
    string inputFileName, outputFileName;
    ifstream inputFile;
    ofstream outputFile;

    map<string, int> wordFrequency;

    set<string> stopWords = {
        "the", "and", "is", "of", "to", "in", "a", "an", "for", "on", "with", "were"
    };

    cout << "Enter input file name: ";
    cin >> inputFileName;

    cout << "Enter output file name: ";
    cin >> outputFileName;

    inputFile.open(inputFileName);

    if (!inputFile) {
        cout << "Error: Could not open input file." << endl;
        return 1;
    }

    string word;

    while (inputFile >> word) {
        word = cleanWord(word);
```

```

int main() {
    while (inputFile >> word) {

        if (word.length() > 3 && stopWords.find(word) == stopWords.end()) {
            wordFrequency[word]++;
        }
    }

    inputFile.close();

    outputFile.open(outputFileName);

    if (!outputFile) {
        cout << "Error: Could not open output file." << endl;
        return 1;
    }

    cout << "\nWords with frequency greater than 2:\n";
    cout << "-----\n";

    outputFile << "Words with frequency greater than 2:\n";
    outputFile << "-----\n";

    for (auto item : wordFrequency) {
        if (item.second > 2) {
            cout << item.first << " : " << item.second << endl;
            outputFile << item.first << " : " << item.second << endl;
        }
    }

    outputFile.close();

```

Launchpad 0 0 Connect You, last we

```

    }

    outputFile.close();

    cout << "\nResults saved successfully to " << outputFileName << endl;

    return 0;
}

```

File to count Words in

Day_02.txtHistory.txt

FileEditViewH1BILSSGridA

Sultanate-e-Usmania (Ottoman Empire)

The Sultanate-e-Usmania, commonly known as the Ottoman Empire, was one of the most powerful and long-lasting empires in world history. It was founded in 1299 CE by Osman I, from whom the empire derived its name. Beginning as a small Turkish state in Anatolia (modern-day Turkey), the Ottoman Empire gradually expanded through military conquests and strategic alliances. Over time, it became a vast empire stretching across Southeast Europe, Western Asia, and North Africa. The Ottomans played a significant role in shaping the political, cultural, and economic history of these regions for more than six centuries.

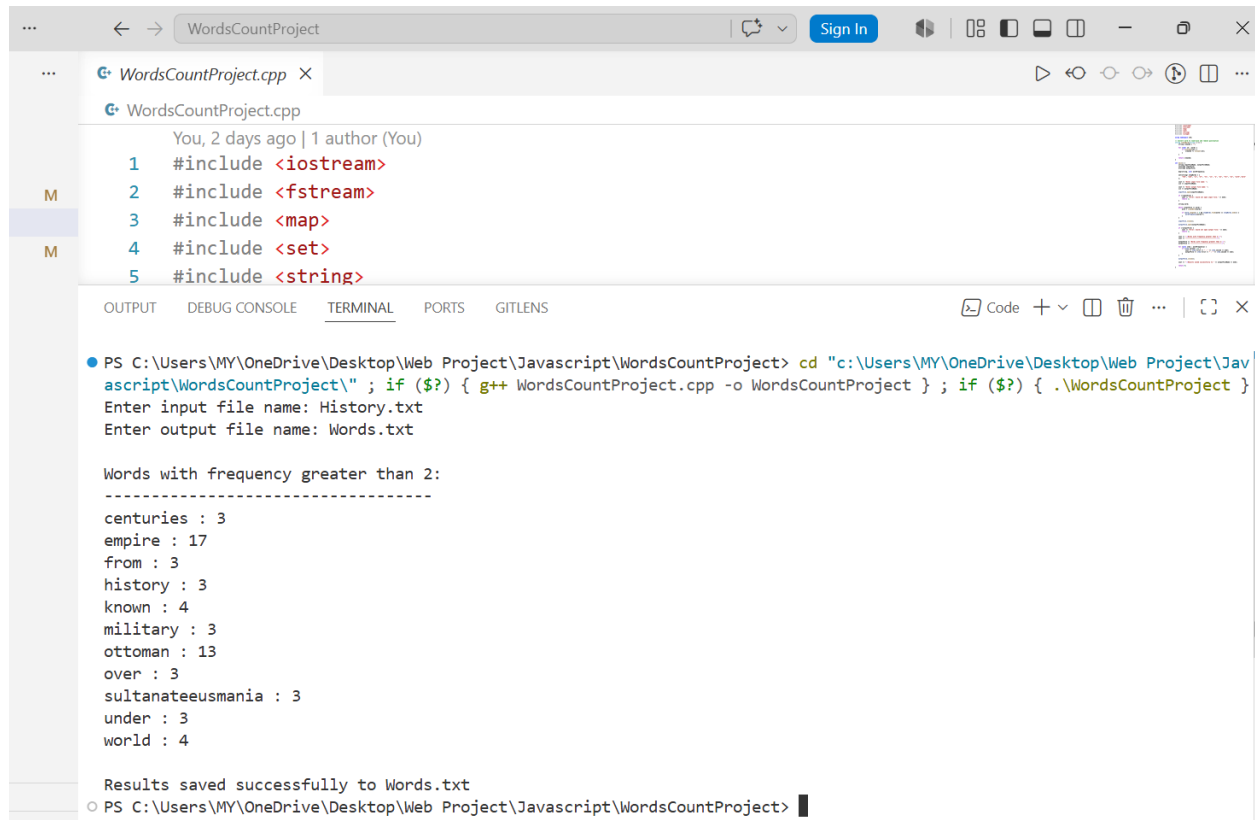
One of the most important events in Ottoman history was the conquest of Constantinople in 1453 by Sultan Mehmed II, also known as Mehmed the Conqueror. This victory ended the Byzantine Empire and transformed Constantinople, later known as Istanbul, into the Ottoman capital. The conquest marked the rise of the Ottomans as a major world power and strengthened their control over important trade routes connecting Europe and Asia. Under successive rulers, the empire continued to expand and reached its greatest territorial extent during the reign of Sultan Suleiman the Magnificent in the sixteenth century.

The Ottoman Empire was known for its effective administrative system and military strength. The empire was divided into provinces governed by appointed officials who ensured law and order throughout the territories. The Ottoman military, especially the elite Janissary Corps, was highly disciplined and played a crucial role in the empire's expansion. The sultans exercised supreme authority, but they were assisted by ministers and advisors who managed state affairs. The empire also maintained a legal system based on Islamic law as well as administrative regulations that addressed the needs of its diverse population.

Culturally, the Ottoman Empire made remarkable contributions to architecture, literature, education, and the arts. Magnificent mosques, palaces, bridges, and public buildings were constructed throughout the empire. Architectural masterpieces such as the Blue Mosque and Topkapı Palace in Istanbul continue to attract visitors from around the world. Ottoman society was multicultural and multi-religious, allowing various ethnic and religious communities to practice their traditions while remaining under Ottoman rule. This diversity contributed to the richness of Ottoman culture and helped the empire maintain stability for many centuries.

Ln 1, Col 13,227 charactersPlain text100%Windows (CRLF)UTF-8

The Result:



The screenshot shows the Visual Studio Code interface with a file named `WordsCountProject.cpp` open. The code is a C++ program that reads words from a file and counts their frequency. The terminal window shows the execution of the program, which prompts for an input file name and an output file name. The output shows the words with their frequency greater than 2.

```
1 #include <iostream>
2 #include <fstream>
3 #include <map>
4 #include <set>
5 #include <string>
```

```
PS C:\Users\MY\OneDrive\Desktop\Web Project\Javascript\WordsCountProject> cd "c:\Users\MY\OneDrive\Desktop\Web Project\JavaScript\WordsCountProject\" ; if ($?) { g++ WordsCountProject.cpp -o WordsCountProject } ; if ($?) { .\WordsCountProject }
Enter input file name: History.txt
Enter output file name: Words.txt

Words with frequency greater than 2:
-----
centuries : 3
empire : 17
from : 3
history : 3
known : 4
military : 3
ottoman : 13
over : 3
sultanateusmania : 3
under : 3
world : 4

Results saved successfully to Words.txt
PS C:\Users\MY\OneDrive\Desktop\Web Project\Javascript\WordsCountProject>
```

Day_02.txt

History.txt

Wo

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```
centuries : 3
empire : 17
history : 3
known : 4
military : 3
ottoman : 13
over : 3
sultanateusmania : 3
under : 3
world : 4
```